```python
# accept 3 sides of triangle and tell its type
a = float(input("Enter 1st Side of the triangle : "))
b = float(input("Enter 2nd Side of the triangle : "))
c = float(input("Enter 3rd Side of the triangle : "))

if(a==b and b==c):
    print("Equilateral Triangle")
elif(a==b or b==c or a==c):
    print("Isoceles Triangle")
else:
    print("Scalene Triangle")
```

```
Enter 1st Side of the triangle : 1
Enter 2nd Side of the triangle : 1
Enter 3rd Side of the triangle : 1
Equilateral Triangle
```

```python
#function solver

print("f(x) = ax^3 - bx^2 + cx -d            if x > k")
print("f(x) = 0                             if x = k")
print("f(x) = -ax^3 + bx^2 - cx +d          if x < k")

a=float(input("Enter value of a : "))
b=float(input("Enter value of b : "))
c=float(input("Enter value of c : "))
d=float(input("Enter value of d : "))
k=float(input("Enter value of k : "))
x=float(input("Enter value of x : "))

if(x>k):
    f = (a*(x**3))-(b*(x**2))+(c*x)-d
    print(f"Solution : {f}")
elif(x==k):
    print("Solution : 0")
else:
    f = (a*(x**3))-(b*(x**2))+(c*x)-d
    print(f"Solution : {f}")
```

```
f(x) = ax^3 - bx^2 + cx -d            if x > k
f(x) = 0                              if x = k
f(x) = -ax^3 + bx^2 - cx +d           if x < k
Enter value of a : 1
Enter value of b : 2
Enter value of c : 3
Enter value of d : 4
Enter value of k : 5
Enter value of x : 6
Solution : 158.0
```

```python
# BMI calculator
w = float(input("Enter weight in kg : "))
h = float(input("Enter height in m : "))

bmi = w/(h**2)

if(bmi <= 18.4):
    print(f"BMI : {bmi:.2f} Status : Underweight")
elif(bmi <= 24.9):
    print(f"BMI : {bmi:.2f} Status : Normal")
elif(bmi <= 39.9):
    print(f"BMI : {bmi:.2f} Status : Overweight")
else:
    print(f"BMI : {bmi:.2f} Status : Obese")
```

```
Enter weight in kg : 105
Enter height in m : 1.83
BMI : 31.35 Status : Overweight
```

```python
#DISTANCE CONVERTER

print("Enter the value of the convereter u want to use")
print("1. Inches to Feet ")
print("2. Centimeter to Meter ")

choice = int(input("Enter the value : "))

if (choice == 1):
    inch = float(input("Enter distance in inches : "))
    feet = inch/12
    print(f"feet : {feet:.2f}")
elif(choice == 2):
    cm = float(input("Enter distance in centimeter : "))
    m = cm/100
    print(f"meter : {m:.2f}m")
else:
    print("Invalid Choice")
```

```
Enter the value of the convereter u want to use
1. Inches to Feet
2. Centimeter to Meter
Enter the value : 2
Enter distance in centimeter : 183
meter : 1.83m
```

```python
#read and display string
s = input("Enter String : ")
print(s)
```

```
Enter String : Hello World1
Hello World1
```

```python
# length of string
s = "Hello World"
print(len(s))
```

```
11
```

```python
# lower case
s = "Python Programming"
s=s.lower()
print(s)
```

```
python programming
```

```python
# concat
s1="Data"
s2="Science"
s = s1 + " " + s2
print(s)
```

```
Data Science
```

```python
#remove trailing spaces
s = "    Data Analysis    "
print(s)
s = s.strip()
print(s)
```

```
    data analysis
data analysis
```

```python
#odd string elements
s ="Data Science"
print(s[1::2])
```

```
aaSine
```

```python
#check if palindrom or not
s = input("Enter String : ")
rs = s[::-1]
if (s == rs ):
    print("Palindrome")
else :
    print("Not Palindrome")
```

```
Enter String : amma
Palindrome
```

```python
#string reverse
s="Hello"
print(s[::-1])
```

```
olleH
```

```python
# the seperator and joiner
s = "Hello world! I am Tired saying this same word"
a = s.split()
sym = input("enter joiner : ")
print(s)
print(a)
print(sym.join(a))
```

```
enter joiner : -
Hello world! I am Tired saying this same word
['Hello', 'world!', 'I', 'am', 'Tired', 'saying', 'this', 'same',
'word']
Hello-world!-I-am-Tired-saying-this-same-word
```

```python
# input words then reverse words order then join by hypen
s = input("Enter the words : ")
a = s.split()
a = a[::-1]
a = '-'.join(a)
print(a)
```

```
Enter the words : Hello world! I am Tired saying this same word
word-same-this-saying-Tired-am-I-world!-Hello
```